

Lecture Notes with Complete Statements and Solutions

Interest Rates and Amortization (Chapter 5)

1 Review of Time Value of Money (TVM)

Previously covered concepts:

- Present Value (PV) and Future Value (FV) of lump sums.
- Solving for missing variable among:

$$PV, FV, i, n$$

- Future Value of Ordinary Annuity and Annuity Due.
- Present Value of Ordinary Annuity.
- Adjusting for non-annual compounding.

2 Effective vs. Nominal Interest Rates

Complete Question

Bank A offers an **8% nominal rate compounded quarterly**. Bank B wants the **same effective annual rate (EAR)** but compounded **monthly**.

Find the nominal rate for Bank B.

Solution

$$EAR_A = \left(1 + \frac{0.08}{4}\right)^4 - 1 = 8.24\%$$

$$0.0824 = \left(1 + \frac{r}{12}\right)^{12} - 1$$

$$r \approx 7.95\%$$

$$\boxed{r \approx 7.95\%}$$

3 Loan Amortization Schedule

Complete Question

A borrower takes a **\$10,000 loan** for **5 years** at **6% annual interest**. Find the payment and construct the full amortization schedule.

Step 1: Annual Payment

$$PMT = 10,000 \times \frac{0.06}{1 - (1.06)^{-5}} \approx 2,373.96$$

$PMT \approx \$2,373.96$

Step 2: Full 5-Year Amortization Table

Year	Opening Balance	Payment	Interest	Principal	Closing Balance
1	10,000.00	2,373.96	600.00	1,773.96	8,226.04
2	8,226.04	2,373.96	493.56	1,880.40	6,345.64
3	6,345.64	2,373.96	380.74	1,993.22	4,352.42
4	4,352.42	2,373.96	261.15	2,112.81	2,239.61
5	2,239.61	2,373.96	134.38	2,239.58	0.00

Key Insight

- Early payments are mostly **interest**.
- Later payments are mostly **principal**.
- Final balance reaches **zero**, confirming correct amortization.

4 Retirement Planning Problem

Complete Question

A person saves **\$8,000 yearly** from age **26 to 65** at **8%**.

1. Find retirement fund at 65.
2. Find equal withdrawals for 20 years.

Future Value at Retirement

$$n = 39$$

$$FV = 8,000 \times \frac{(1.08)^{39} - 1}{0.08} \approx 3,021,582$$

$$\boxed{FV \approx \$3,021,582}$$

Annual Withdrawal for 20 Years

$$PMT = 3,021,582 \times \frac{0.08}{1 - (1.08)^{-20}} \approx 307,700$$

$$\boxed{\text{Annual Withdrawal} \approx \$307,700}$$

5 Key Exam Insights

- Convert nominal rates to EAR before comparison.
- Loan payments use PV of ordinary annuity.
- Retirement problems have:
 - Saving phase (future value).
 - Withdrawal phase (present value).

6 Quiz Reminder

$\boxed{\text{TVM Quiz at 4:00 PM next class}}$